

EXPERIMENT NO. 16:

Compare different types of Classification Algorithms and evaluate their performance.

Aim:

To implement and compare different Classification Algorithms in Python and evaluate their performance using accuracy.

Procedure:

1. Import the required Python libraries.
2. Load the Iris dataset.
3. Split the dataset into training and testing sets.
4. Implement different classification algorithms such as KNN, Naive Bayes, Logistic Regression, and Decision Tree.
5. Train each classifier using the training dataset.
6. Predict the classes using the testing dataset.
7. Calculate the accuracy of each algorithm.
8. Compare the performance of all classification algorithms.

Program:

```
# Comparison of Classification Algorithms

from sklearn.datasets import load_iris

from sklearn.model_selection import train_test_split

from sklearn.neighbors import KNeighborsClassifier

from sklearn.naive_bayes import GaussianNB

from sklearn.linear_model import LogisticRegression

from sklearn.tree import DecisionTreeClassifier

from sklearn.metrics import accuracy_score

# Load Iris Dataset

iris = load_iris()

X = iris.data

y = iris.target

# Split dataset

X_train, X_test, y_train, y_test = train_test_split(

    X, y, test_size=0.3, random_state=42

)
```

```

# Create Classification Models

models = {

    "KNN": KNeighborsClassifier(n_neighbors=3),

    "Naive Bayes": GaussianNB(),

    "Logistic Regression": LogisticRegression(max_iter=200),

    "Decision Tree": DecisionTreeClassifier(random_state=42)

}

# Train and evaluate each model

print("Classification Algorithm Performance:\n")

for name, model in models.items():

    model.fit(X_train, y_train)

    y_pred = model.predict(X_test)

    accuracy = accuracy_score(y_test, y_pred)

    print(name)

    print("Accuracy:", round(accuracy * 100, 2), "%")

    print()

```

Output:

```

... Classification Algorithm Performance:

KNN
Accuracy: 100.0 %

Naive Bayes
Accuracy: 97.78 %

Logistic Regression
Accuracy: 100.0 %

Decision Tree
Accuracy: 100.0 %

```

Result:

The different Classification Algorithms were successfully implemented and compared using the Iris dataset. KNN, Logistic Regression, and Decision Tree achieved the highest accuracy for this particular train-test split, while Naive Bayes also achieved high accuracy. Thus, the performance of different classification algorithms can be evaluated and compared using classification accuracy.